

Nneka Hamilton

CS 499

Artifact Two: Algorithms and Data Structures Narrative

Data Structures and Algorithms.

I opted to enhance the Algorithms and Data Structures with Traveler Gateway to improve the application of data handling and display. The first prototype enabled the user to choose from a number of different trip alternatives, but the user could not search, sort or assess these options.

Enhancement

The most important change was the introduction of reusable algorithms for the optimization of processing trip information. I wrote an algorithm that allows users to search for trips by location, trip code, difficulty rating, or travel categories.

I can filter flights by price, duration, user rating, or alphabetically. Instead, then writing separate functions for each sorting algorithm, I created a simple sorting function that chooses the algorithm based on the user's input.

To help optimize my data, I designed algorithms to calculate travel statistics such as the total number of journeys, average cost of a trip, and the most and least expensive places. I built a feature location function that highlights the most often taken trip in the program.

Efficient JavaScript array methods like `filter()`, `sort()`, `reduce()` and `some()` can make it easier to make data modifications in apps. These strategies allow us to avoid duplicating logic.

The act of thinking

This modification demonstrated the effect of algorithms on the performance of the software and the happiness of its users. Helper functions can encapsulate reusable algorithms that perform specific instructions, making the software more maintainable, readable, and scalable.

These advances demonstrated the importance of data structures and algorithms for solving real software engineering problems.